Classes of PDEs:

- Elliptic: $\nabla^2 u = f(\vec{x},t)$
- Parabolic: $u_t = \nabla^2 u + f(\vec{x},t)$
- Hyperbolic: $u_{tt} = c^2 \nabla^2 u$

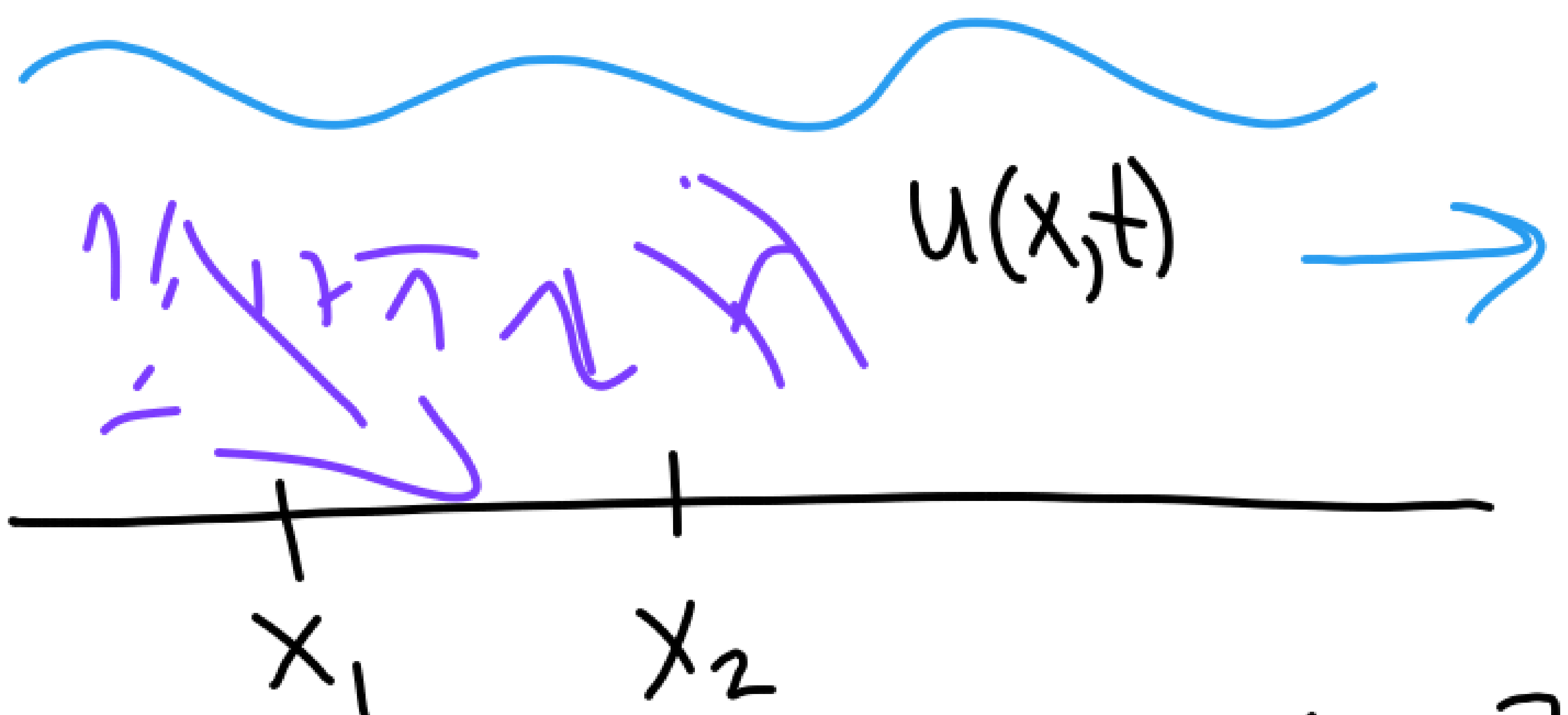
  Usually written in 1st-order form

Hyperbolic PDEs

model waves:

- Sound
- Electromagnetic
- Fluid dynamics
- Surface water waves

Flow of a fluid in a narrow channel



Total amount of dye in $[x_1, x_2]$:

$$\int_{x_1}^{x_2} u(x,t)\,dx$$

This changes in time due to flux through the endpoints:

$$\frac{d}{dt}\int_{x_1}^{x_2} u\,dx = f(u(x_1,t)) - f(u(x_2,t))$$

If u is smooth enough:

$$\int_{x_1}^{x_2} \frac{\partial}{\partial t} u(x,t)\,dx = -\int_{x_1}^{x_2} \frac{\partial}{\partial x} f(u(x,t))\,dx$$

$$\int_{x_1}^{x_2} \left(u_t + f(u)_x\right) dx = 0$$

Must hold on every interval $[x_1, x_2]$

So the integrand must vanish pointwise:

$$u_t + f(u)_x = 0$$

Let's take the simplest case: constant velocity a

$$f(u) = au$$

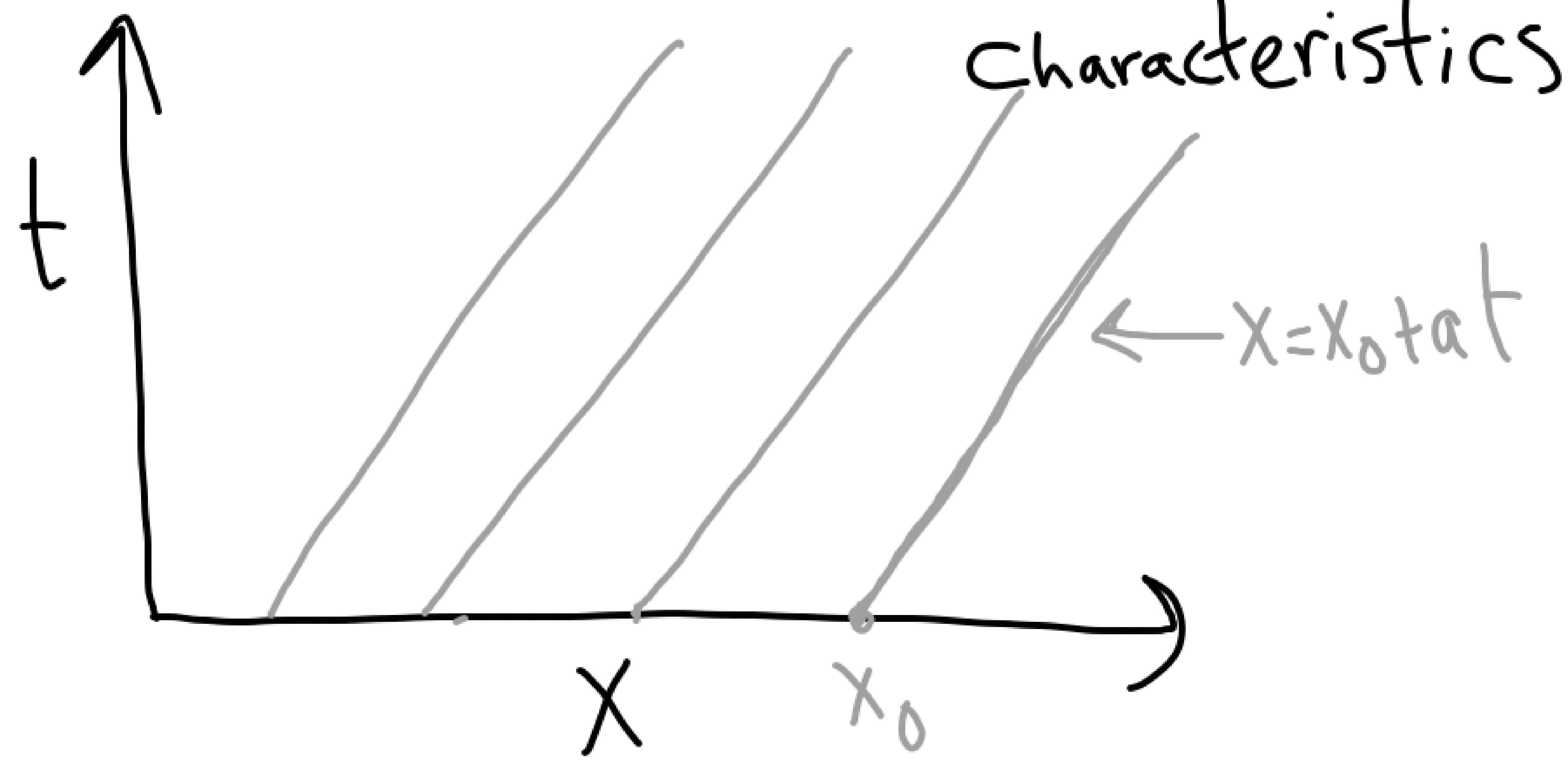
$$u_t + au_x = 0$$ Advection equation

$$u(x, t=0) = \eta(x)$$ Initial data

Cauchy problem: $x \in \mathbb{R}$

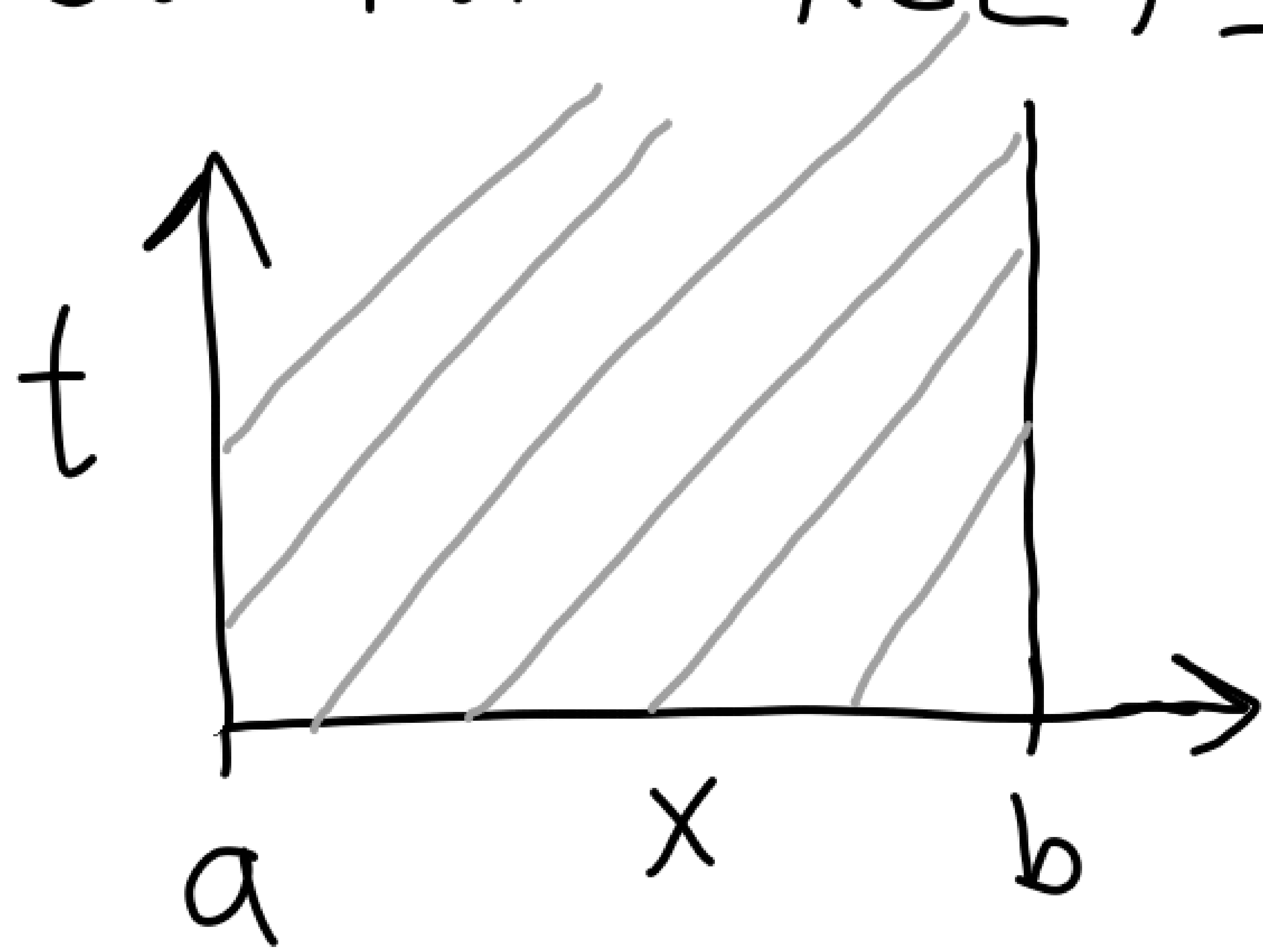
Solution: $u(x,t) = \eta(x-at)$

Check: $-a\eta' + a\eta' = 0$



Soln. is constant along each characteristic

What about bounded domains? $x \in [a,b]$



What boundary conditions will determine a unique solution?

$u(x=a,t) = \alpha(t)$ (if $a>0$) or $u(x=b,t) = \beta(t)$ (if $a<0$)

Discretization

Centered in space, forward in time

$$\frac{U_j^{n+1} - U_j^n}{k} + a\frac{U_{j+1}^n - U_{j-1}^n}{2h} = 0$$

$$U_j^{n+1} = U_j^n - \frac{ka}{2h}\left(U_{j+1}^n - U_{j-1}^n\right)$$

1st order in time
2nd order in space

Is it stable?

Von Neumann analysis

$e^{\pm i\theta} = \cos\theta \pm i\sin\theta$

$$U_j^n \to g^n e^{ijh\xi}$$

$$g^{n+1} e^{ijh\xi} = g^n e^{ijh\xi} - \frac{ka}{2h} g^n \left(e^{i(j+1)h\xi} - e^{i(j-1)h\xi}\right)$$

$$g = 1 - \frac{ka}{2h}\left(e^{ih\xi} - e^{-ih\xi}\right)$$

$$= 1 - \frac{ka}{2h}\left(2i\sin(h\xi)\right)$$

$$g = 1 - \frac{ka}{h} i\sin(h\xi)$$

$$|g|^2 = \left(\mathrm{Re}(g)\right)^2 + \left(\mathrm{Im}(g)\right)^2$$

$$= 1 + \frac{k^2 a^2}{h^2}\sin^2(h\xi) > 1$$

Not stable!

Why this ansatz?

Weierstrass: $u(x)$ can be approximated by $\int \hat{u}(\xi) e^{i\xi x}\, d\xi$

Discrete analog:
U can be represented by $\sum \hat{U}_j e^{ijh\xi}$ (i.e. they form a basis)
and they are the eigenvectors of

↗ circulant matrices.

Method of lines analysis

Discretize first in space:

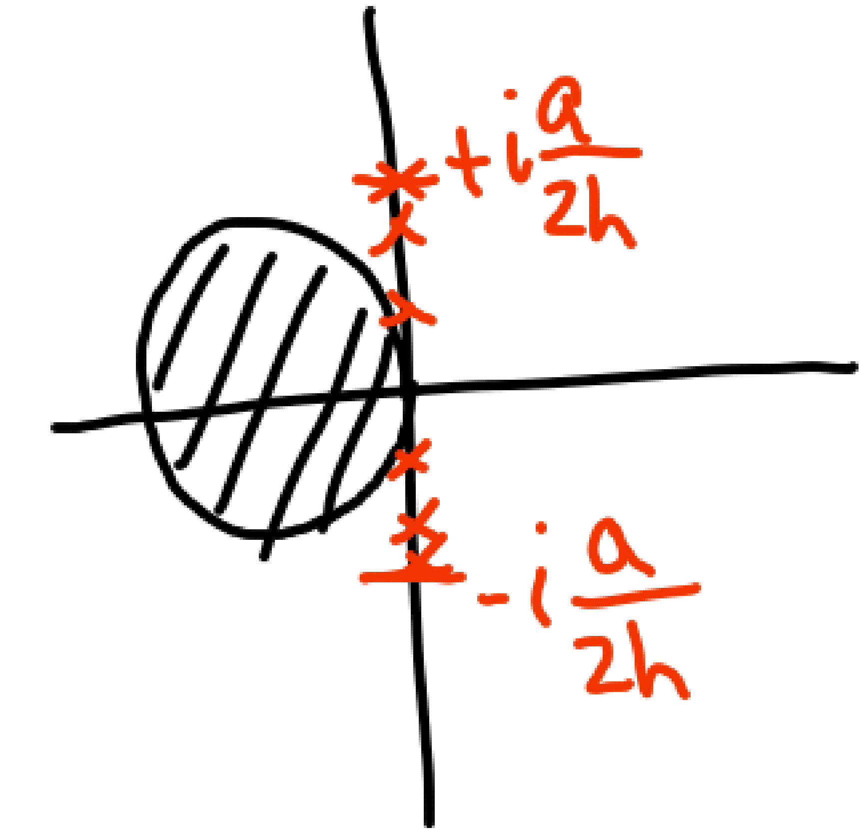
$$U_j'(t) = -\frac{a}{2h}\left(U_{j+1} - U_{j-1}\right)$$

$$U'(t) = -\frac{a}{2h}\begin{bmatrix} 0 & 1 & & & \\ -1 & \ddots & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & -1 & 0 \end{bmatrix}$$

$A^T = -A$ skew-symmetric $\Rightarrow$ imaginary eigenvalues

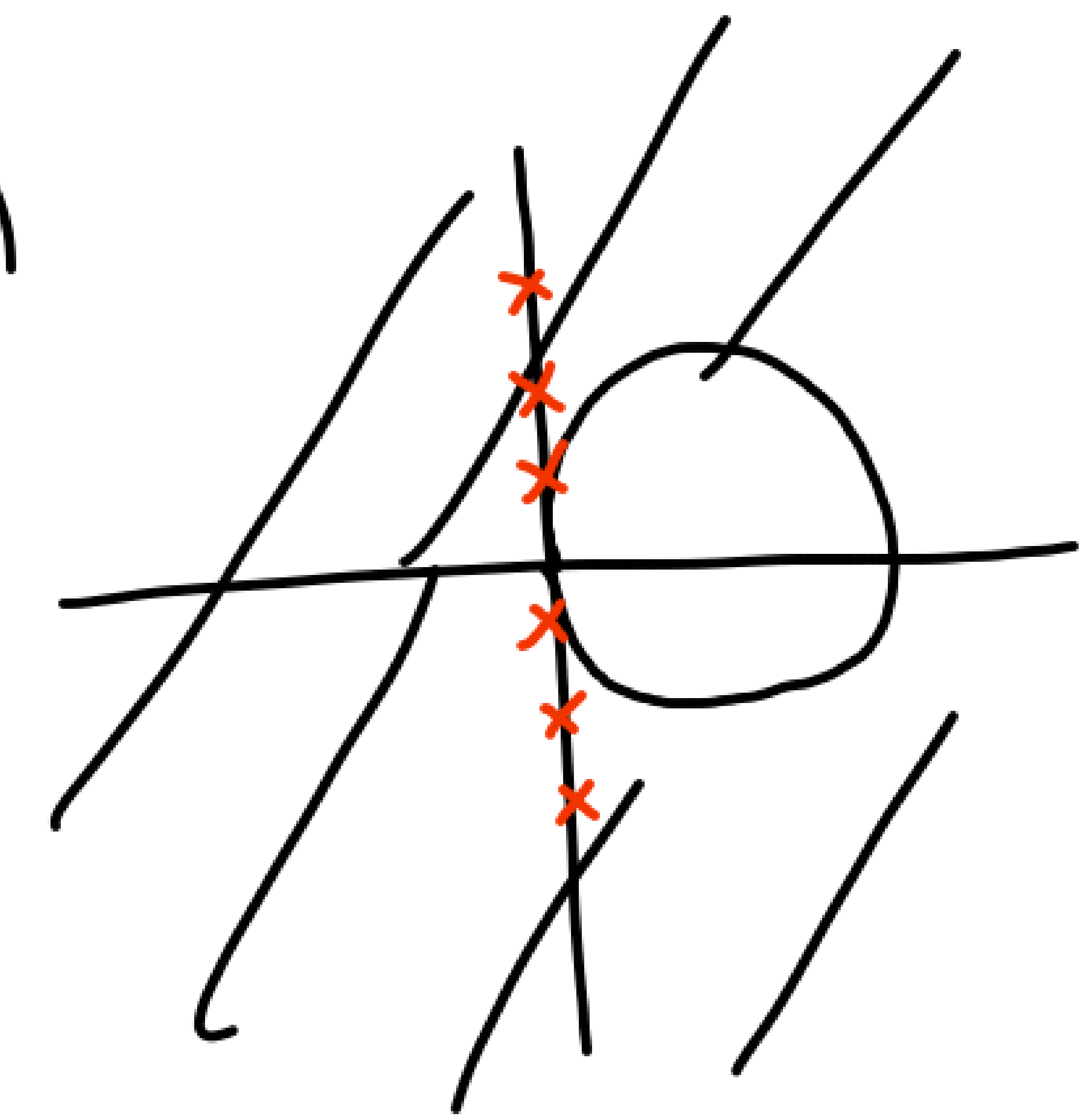
Explicit Euler:

exp. growth



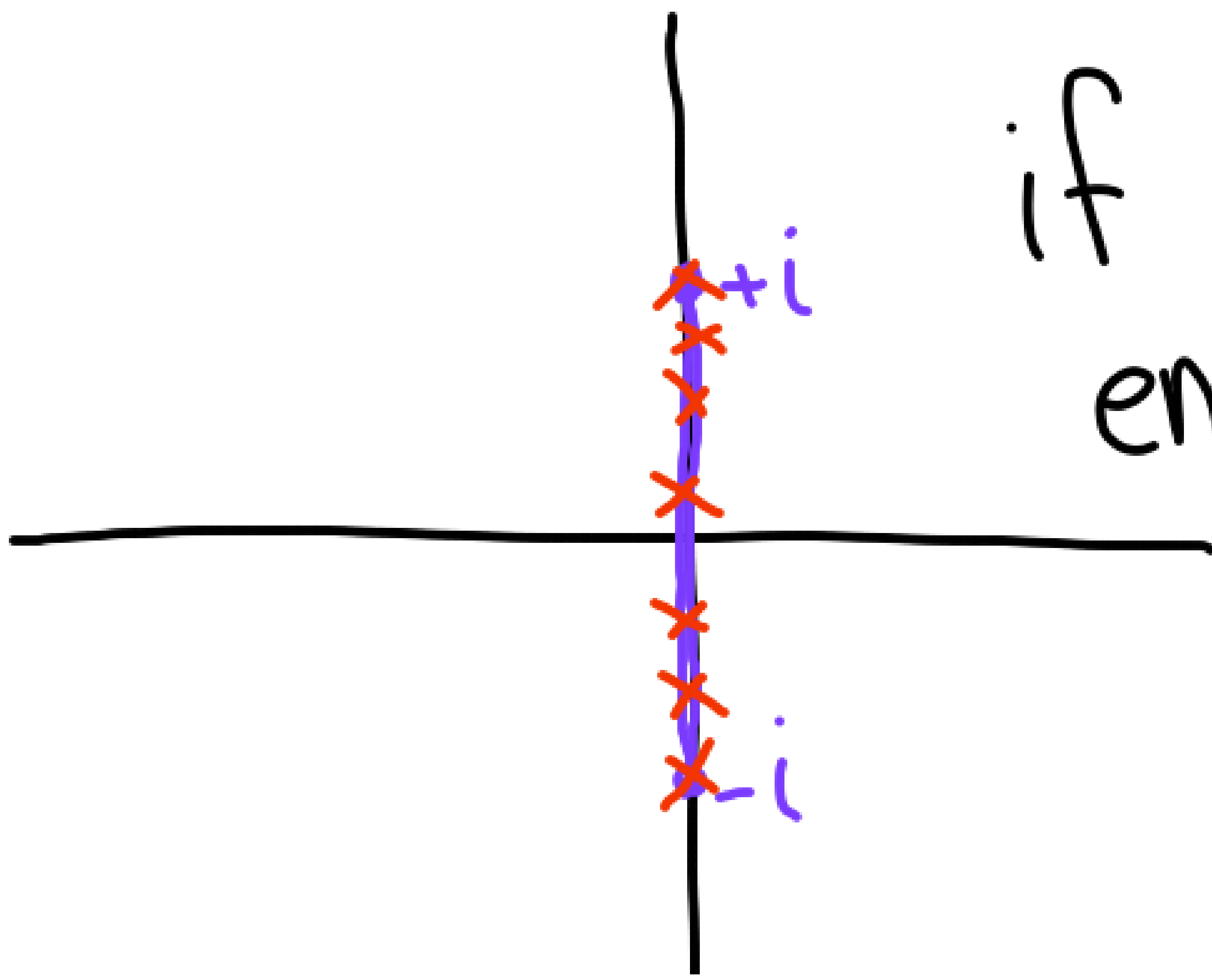
Backward Euler:

exp. decay



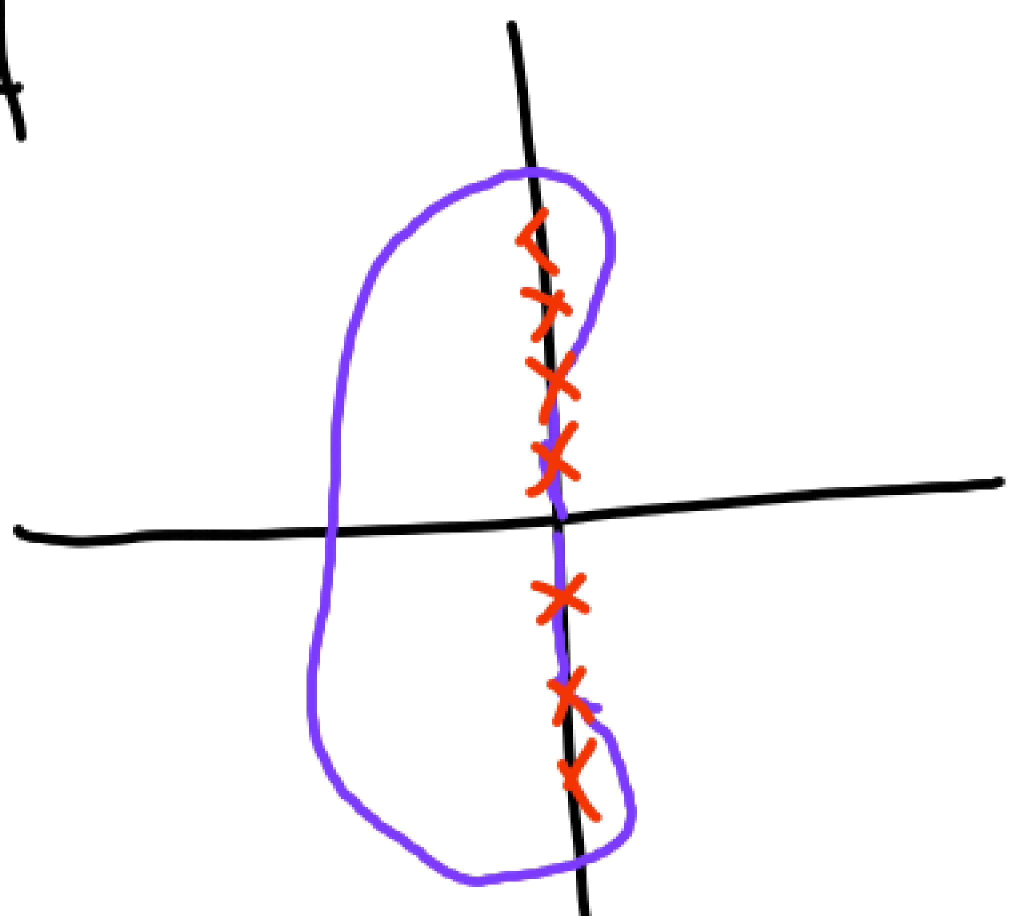
Leapfrog:

$$U_j^{n+1} = U_j^{n-1} - \frac{ka}{2h}\left(U_{j+1}^n - U_{j-1}^n\right)$$



if k is small enough $\left(\frac{ka}{h} \le 1\right)$

RK4



Lax-Friedrichs

$$U_j^{n+1} = \frac{U_{j+1}^n + U_{j-1}^n}{2} - \frac{ka}{2h}\left(U_{j+1}^n - U_{j-1}^n\right)$$

Observe:

$$\frac{U_{j+1}^n + U_{j-1}^n}{2} = U_j^n + \frac{1}{2}\underbrace{\left[U_{j+1}^n - 2U_j^n + U_{j-1}^n\right]}_{\approx h^2 u''(x_j)}$$

$$\frac{U_j^{n+1} - U_j^n}{k} = \frac{h^2}{2k}\,\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2} - \frac{a}{2h}\left(U_{j+1}^n - U_{j-1}^n\right)$$

$$\Rightarrow u_t = \frac{h^2}{2k}u_{xx} - a u_x + \mathcal{O}(k, h^2)$$

So we're approximating

$$u_t + a u_x = \underbrace{\frac{h^2}{2k} u_{xx}}_{\text{Diffusive term}}$$

This is especially useful for nonlinear hyperbolic PDEs.